

**STANDARD OPERATING PROCEDURE
CALIBRATION OF FIELD INSTRUMENTS**
(temperature, pH, dissolved oxygen, conductivity/specific conductance,
oxidation/reduction potential [ORP], and turbidity)

Quality Assurance Unit
U.S. Environmental Protection Agency – Region 1
11 Technology Drive
North Chelmsford, MA 01863

The controlled version of this document is the electronic version viewed on-line only. If this is a printed copy of the document, it is an uncontrolled version and may or may not be the version currently in use.

This document contains direction developed solely to provide guidance to U.S. Environmental Protection Agency (EPA) personnel. EPA retains the discretion to adopt approaches that differ from these procedures on a case-by-case basis. The procedures set forth do not create any rights, substantive or procedural, enforceable at law by party to litigation with EPA or the United States.

Prepared by: Charles Porfett 1/19/10
(Charles Porfett, Quality Assurance Unit) Date

Approved by: Gerard Sotolongo 1-19-10
(Gerard Sotolongo, Quality Assurance Unit) Date

Revision Page

[illegible]

TABLE OF CONTENTS

1.0	SCOPE AND APPLICATION.....	4
2.0	HEALTH AND SAFETY WARNINGS	4
3.0	GENERAL	4
4.0	FREQUENCY OF CALIBRATION	5
5.0	CALIBRATION PROCEDURES	5
5.1	TEMPERATURE	6
5.2	pH (electrometric)	7
5.3	DISSOLVED OXYGEN.	8
5.4	SPECIFIC CONDUCTANCE	10
5.5	OXIDATION/REDUCTION POTENTIAL (ORP)	11
5.6	TURBIDITY	12
6.0	POST CALIBRATION CHECK	14
7.0	DATA MANAGEMENT AND RECORDS MANAGEMENT	15
8.0	REFERENCES.....	15
	Table 1 INSTRUMENT CALIBRATION LOG	16
	OXYGEN SOLUBILITY.....	17

1.0 SCOPE AND APPLICATION

The purpose of this standard operating procedure (SOP) is to provide a framework for calibrating field instruments used to measure water quality parameters for groundwater and surface water. Water quality parameters include temperature, pH, dissolved oxygen, specific conductance, oxidation/reduction potential [ORP], and turbidity. This SOP supplements, but does not replace, EPA analytical methods listed in 40 CFR 136 and 40 CFR 141 for temperature, dissolved oxygen, conductivity/specific conductance, pH and turbidity.

This SOP is written for instruments that measure temperature, pH, dissolved oxygen, specific conductance, turbidity, and/or oxidation/reduction potential [ORP] and the probe readings for pH, dissolved oxygen, and specific conductance are automatically corrected for temperature.

For groundwater monitoring, the instrument must be equipped with a flow-through-cell and the display/logger or computer display screen needs to be large enough to simultaneously contain the readouts of each probe in the instrument. Turbidity is measured using a separate instrument. It must not be measured in a flow-through-cell because the flow-through-cell acts as a sediment trap. This procedure is applicable for use with the *EPA Region 1 Low Stress (low flow) Purging and Sampling Procedure for the Collection of Ground Water Samples from Monitoring Wells*.

2.0 HEALTH AND SAFETY WARNINGS

Read all labels on the standards and note any warnings on the labels. Wear appropriate personal protection equipment (e.g., gloves, eye shields, etc.) when handling the standards. If necessary, consult the Material Safety Data Sheets (MSDS) for additional safety information on the chemicals in the standards.

3.0 GENERAL

All monitoring instruments must be calibrated before they are used to measure environmental samples. For instrument probes that rely on the temperature sensor (pH, dissolved oxygen, specific conductance, and oxidation/reduction potential [ORP]), each temperature sensor needs to be checked for accuracy against a thermometer that is traceable to the National Institute of Standards and Technology (NIST). Before any instrument is calibrated or used to perform environmental measurements, the instrument must stabilize (warm-up) according to manufacturer's instructions and must have no air bubbles lodged between the probe and probe guard.

Most projects will require at least two standards to bracket the expected measurement range. This means that one standard is less than the expected value and one is higher. When an environmental sample measurement falls outside the calibration range, the instrument must be re-calibrated to bracket the new range before continuing measurements. Otherwise, the measurements that are outside the calibration range will need to be qualified.

This SOP requires that the manufacturer's instruction manual (including the instrument specifications) accompany the instrument into the field.

4.0 FREQUENCY OF CALIBRATION

At a minimum, the instrument is calibrated prior to use on the day the measurements are to be performed. A post calibration check at the end of the day is performed to determine if the instrument drifted out of calibration. Some projects may require more frequent calibration checks throughout the day in addition to the check at the end of the day. For these checks, the instrument can be recalibrated during the day if the instrument drifted out of calibration and only the data measured prior to the check would need to be qualified. The calibration/post calibration data information is recorded in Table 1.

Instruments (e.g., sonde) that monitor continuously over a period of time are calibrated before deployment. When these instruments are recovered, the calibration is checked to determine if any of them drifted out of calibration.

Some instruments lose their calibration criteria when they are turned off. Those instruments can either be left on all day (battery dependent) or calibrated at each sampling location. If they are calibrated at each sampling location, a post calibration check is not needed.

Ideally, the temperature of the standards should be close to the temperature of the ambient water that is being measured.

5.0 CALIBRATION PROCEDURES

Prior to calibration, all instrument probes and cable connections must be cleaned and the battery checked according to the manufacturer's instructions. Failure to perform these steps (proper maintenance) can lead to erratic measurements.

If a multi-probe instrument is to be used, program the instrument to display the parameters to be measured (e.g., temperature, pH, percent dissolved oxygen, mg/L dissolved oxygen, specific conductance, and ORP).

The volume of the calibration solutions must be sufficient to cover both the probe and temperature sensor (see manufacturer's instructions for the volume to be used).

Check the expiration date of the standards. Do not use expired standards.

All standards are stored according to manufacturer instructions.

5.1 TEMPERATURE

Most instrument manuals state there is no calibration of the temperature sensor, but the temperature sensor must be checked to determine its accuracy. This accuracy check is performed at least once per year and the accuracy check date/information is kept with the instrument. If the accuracy check date/information is not included with the instrument or the last check was over a year, the temperature sensor accuracy needs to be checked at the beginning of the sampling event. If the instrument contains multiple temperature sensors, each sensor must be checked. This procedure is not normally performed in the field. If the instrument is obtained from a rental company, the rental company should perform the calibration check and include with the instrument documentation that it was performed.

Verification Procedure

1. Fill a container with water and adjust the water temperature to below the water body's temperature to be measured. Use ice or warm water to adjust the temperature.
2. Place a thermometer that is traceable to the National Institute of Standards and Technology (NIST) and the instrument's temperature sensor into the water. Wait for both temperature readings to stabilize.
3. Compare the two measurements. The instrument's temperature sensor must agree with the reference thermometer measurement within the accuracy of the sensor (e.g., $\pm 0.2^{\circ}\text{C}$). If the measurements do not agree, the instrument may not be working properly and the manufacturer needs to be consulted.
4. Adjust the water temperature to a temperature higher than the water body to be measured.
5. Compare the two measurements. The instrument's temperature sensor must agree with the reference thermometer measurement within the accuracy of the sensor (e.g.,

$\pm 0.2^{\circ}\text{C}$). If the measurements do not agree, the instrument may not be working properly and the manufacturer needs to be consulted.

5.2 pH (electrometric)

The pH of a sample is determined electrometrically using a glass electrode.

Choose the appropriate buffered standards that will bracket the expected values at the sampling locations. If the water body's pH is unknown, then three standards are needed for the calibration: one close to seven, one at least two pH units below seven, and the other at least two pH units above seven. Instruments that will not accept three standards will need to be re-calibrated if the water sample's pH is outside the initial calibration range described by the two standards.

Calibration Procedure

1. Allow the buffered standards to equilibrate to the ambient temperature.
2. Fill calibration containers with the buffered standards so each standard will cover the pH probe and temperature sensor.
3. Remove probe from its storage container, rinse with deionized water, and remove excess water.
4. Select measurement mode. Immerse probe into the initial standard (e.g., pH 7).
5. Wait until the readings stabilize. If the reading does not change within 30 seconds, select calibration mode and then select "pH". Enter the buffered standard value into instrument.
6. Remove probe from the initial standard, rinse with deionized water, and remove excess water.
7. Immerse probe into the second standard (e.g., pH 4). Repeat step 5.
8. Remove probe from the second standard, rinse with deionized water, and remove excess water. If instrument only accepts two standards, the calibration is complete. Go to step 11. Otherwise continue.
9. Immerse probe in third buffered standard (e.g., pH 10) and repeat step 5.

10. Remove probe from the third standard, rinse with deionized water, and remove excess water.
11. Select measurement mode, if not already selected. To ensure that the initial calibration standard (e.g., pH 7) has not changed, immerse the probe into the initial standard. Wait for the readings to stabilize. The reading should read the initial standard value within the manufacturer's specifications. If not, re-calibrate the instrument. If re-calibration does not help, the calibration range may be too great. Reduce calibration range by using standards that are closer together.
12. The calibration is complete. Rinse the probe with deionized water and store the probe according to manufacturer's instructions.
13. Record the calibration information on Table 1.

5.3 DISSOLVED OXYGEN

Dissolved oxygen (DO) content in water is measured using a membrane electrode. To insure proper operation, the DO probe's membrane and electrolyte should be replaced prior to calibration for the sampling event. The new membrane may need to be conditioned before it is used; consult manufacturer's manual on how the conditioning is to be performed. Failure to perform this step may lead to erratic measurements. Before performing the calibration/measurements, inspect the membrane for air bubbles and nicks.

Note: some manufacturers require an altitude correction instead of a barometric correction. In that case, enter the altitude correction according to the manufacturer's directions in Step 5 and then proceed to Step 6.

Note: some instruments have a built-in barometer. Follow the manufacturer's instructions for entering the barometric value in step 5.

Calibration Procedure

1. Gently dry the temperature sensor and remove any water droplets from the DO probe's sensor membrane according to manufacturer's instructions. Note that the evaporation of moisture on the temperature sensor or DO probe may influence the readings during calibration.
2. Create a 100 percent water-saturated air environment by placing a wet sponge or a

wet paper towel on the bottom of the DO calibration container. Place the DO probe into the calibration container. The probe is loosely fitted into the calibration container to prevent the escape of moisture evaporating from the sponge or paper towel while maintaining ambient pressure (see manufacturer's instructions). Note that the probe and the temperature sensor must not come in contact with these wet items.

3. Allow the confined air to become saturated with water vapor (saturation occurs in approximately 10 to 15 minutes). During this time, turn on the instrument to allow the DO probe to warm-up. Select the measurement mode. Check the temperature readings. Readings must stabilize before continuing to the next step.
4. Select calibration mode; then select "DO %".
5. Enter the local barometric pressure (usually in mm of mercury) for the sampling location into the instrument. This measurement must be determined from an on-site barometer. Do not use barometric pressure obtained from the local weather services unless the pressure is corrected for the elevation of the sampling location. [Note: inches of mercury times 25.4 mm/inch equals mm of mercury or consult Oxygen Solubility at Indicated Pressure chart attached to the SOP for conversion at selected pressures].
6. The instrument should indicate that the calibration is in progress. After calibration, the instrument should display percent saturated DO.
7. Select measurement mode and set the display to read DO mg/L and temperature. Compare the DO mg/L reading to the Oxygen Solubility at Indicated Pressure chart attached to the SOP. The numbers should agree. If they do not agree within the accuracy of the instrument (usually ± 0.2 mg/L), repeat calibration. If this does not work, change the membrane and electrolyte solution.
8. Remove the probe from the container and place it into a 0.0 mg/L DO solution (see footnote). Check temperature readings. They must stabilize before continuing.
9. Wait until the "mg/L DO" readings have stabilized. The instrument should read less than 0.5 mg/L (assuming an accuracy of ± 0.2 mg/L). If the instrument reads above 0.5 mg/L or reads negative, it will be necessary to clean the probe, and change the membrane and electrolyte solution. If this does not work, try a new 0.0 mg/L DO solution. If these changes do not work, contact the manufacturer. Note: some projects and instruments may have different accuracy requirements. The 0.5 mg/L

value may need to be adjusted based on the accuracy requirements of the project or instrument.

10. After the calibration has been completed, rinse the probe with tap or deionized water and store the probe according to manufacturer's instructions. It is important that all of the 0.0 mg/L DO solution be rinsed off the probe so as not to effect the measurement of environmental samples.
11. Record calibration information on Table 1.

Note: You can either purchase the 0.0 mg/L DO solution from a vendor or prepare the solution yourself. To prepare a 0.0 mg/L DO solution, follow the procedure stated in Standard Methods (Method 4500-O G). The method basically states to add excess sodium sulfite (until no more dissolves) and a trace amount of cobalt chloride (read warning on the label before use) to water. This solution is prepared prior to the sampling event. Note: this solution can be made without cobalt chloride, but the probe will take longer to respond to the low DO concentration.

5.4 SPECIFIC CONDUCTANCE

Conductivity is used to measure the ability of an aqueous solution to carry an electrical current. Specific conductance is the conductivity value corrected to 25°C.

Most instruments are calibrated against a single standard which is near the specific conductance of the environmental samples. The standard can be either below or above the specific conductance of the environmental samples. A second standard is used to check the linearity of the instrument in the range of measurements.

When performing specific conductance measurement on groundwater or surface water and the measurement is outside the initial calibration range defined by the two standards, the instrument will need to be re-calibrated using the appropriate standards.

Specific Conductance Calibration Procedure

1. Allow the calibration standards to equilibrate to the ambient temperature.
2. Fill calibration containers with the standards so each standard will cover the probe and temperature sensor. Remove probe from its storage container, rinse the probe with deionized water or a small amount of the standard (discard the rinsate), and place the

probe into the standard.

3. Select measurement mode. Wait until the probe temperature has stabilized.
4. Select calibration mode, then specific conductance. Enter the specific conductance standard value. Make sure that the units on the standard are the same as the units used by the instrument. If not, convert the units on the standard to the units used by the instrument.
5. Select measurement mode. The reading should remain within manufacturer's specifications. If it does not, re-calibrate. If readings continue to change after re-calibration, consult manufacturer or replace calibration solution.
6. Remove probe from the standard, rinse the probe with deionized water or a small amount of the second standard (discard the rinsate), and place the probe into the second standard. The second standard will serve to verify the linearity of the instrument. Read the specific conductance value from the instrument and compare the value to the specific conductance on the standard. The two values should agree within the specifications of the instrument. If they do not agree, re-calibrate. If readings do not compare, then the second standard may be outside the linear range of the instrument. Use a standard that is closer to the first standard and repeat the verification. If values still do not compare, try cleaning the probe or consult the manufacturer.
7. After the calibration has been completed, rinse the probe with deionized water and store the probe according to manufacturer's instructions.
8. Record the calibration information on Table 1.

Note: for projects where specific conductance is not a critical measurement it may be possible to calibrate with one standard in the range of the expected measurement.

5.5 OXIDATION/REDUCTION POTENTIAL (ORP)

The oxidation/reduction potential is the electrometric difference measured in a solution between an inert indicator electrode and a suitable reference electrode. The electrometric difference is measured in millivolts and is temperature dependent.

Calibration or Verification Procedure

1. Allow the calibration standard (a Zobell solution: read the warning on the label before use) to equilibrate to ambient temperature.
2. Remove the probe from its storage container and place it into the standard.
3. Select measurement mode.
4. Wait for the probe temperature to stabilize, and then read the temperature.
5. If the instrument is to be calibrated, do Steps 6 and 7. If the instrument calibration is to be verified, then go to Step 8.
6. Look up the millivolt (mv) value at this temperature from the millivolt versus temperature correction table usually found on the standard bottle or on the standard instruction sheet. You may need to interpolate millivolt value between temperatures. Select "calibration mode", then "ORP". Enter the temperature-corrected ORP value into the instrument.
7. Select measurement mode. The readings should remain unchanged within manufacturer's specifications. If they change, re-calibrate. If readings continue to change after re-calibration, try a new Zobell solution or consult manufacturer. Go to Step 9.
8. If the instrument instruction manual states that the instrument is factory calibrated, then verify the factory calibration against the Zobell solution. If they do not agree within the specifications of the instrument, try a new Zobell solution. If it does not agree, the instrument will need to be re-calibrated by the manufacturer.
9. After the calibration has been completed, rinse the probe with deionized water and store the probe according to manufacturer's instructions.
10. Record the calibration information on Table 1.

5.6 TURBIDITY

The turbidity method is based upon a comparison of intensity of light scattered by a sample under defined conditions with the intensity of light scattered by a standard reference suspension. A

turbidimeter is a nephelometer with a visible light source for illuminating the sample and one or more photo-electric detectors placed ninety degrees to the path of the light source. Note: the below calibration procedure is for a turbidimeter which the sample is placed into a cuvette.

Some instruments will only accept one standard. For those instruments, the second, third, etc., standards will serve as check points.

Calibration Procedures

1. Allow the calibration standards to equilibrate at the ambient temperature. The use of commercially available polymer primary standards (AMCO-AEPA-1) is preferred; however, the standards can be prepared using Formazin (read the warning on the label before use) according to the EPA analytical Method 180.1. Other standards may be used if they can be shown that they are equivalent to the previously mentioned standards.
2. If the standard cuvette is not sealed, rinse a cuvette with deionized water. Shake the cuvette to remove as much water as possible. Do not wipe dry the inside of the cuvette because lint from the wipe may remain in the cuvette. Add the standard to the cuvette.
3. Before performing the calibration procedure, make sure the cuvettes are not scratched and the outside surfaces are dry and free from fingerprints and dust. If the cuvette is scratched or dirty, discard or clean the cuvette respectively. Note: some manufacturers require the cuvette to be orientated in the instrument in a particular direction for accurate reading.
4. Select a low value standard such as a zero or 0.02 NTU and calibrate according to manufacturer's instructions. Note: a zero standard (approximately 0 NTU) can be prepared by passing distilled water through a 0.45 micron pore size membrane filter.
5. Select a high standard and calibrate according to manufacturer's instructions or verify the calibration if instrument will not accept a second standard. In verifying, the instrument should read the standard value to within the specifications of the instrument. If the instrument has range of scales, check each range that will be used during the sampling event with a standard that falls within that range.
6. Record the calibration information on Table 1.

6.0 POST CALIBRATION CHECK

After the initial calibration is performed, the instrument's calibration may drift during the measurement period. As a result, you need to determine the amount of drift that occurred after collecting the measurements. This is performed by placing the instrument in measurement mode (not calibration mode) and placing the probe in one or more of the standards used during the initial calibration; for turbidity place the standard in a cuvette and then into the turbidimeter. Wait for the instrument to stabilize and record the measurement (Table 1). Compare the measurement value to the initial calibration value. This difference in value is then compared to the drift criteria or post calibration criteria described in the quality assurance project plan or the sampling and analysis plan for the project. If the check value is outside the criteria, then the measurement data will need to be qualified.

For the dissolved oxygen calibration check, follow the calibration instructions steps one through three while the instrument is in measurement mode. Record dissolved oxygen value (mg/L), temperature, and barometric pressure. Compare the measurement value to the Oxygen Solubility at Indicated Pressure chart attached to this SOP. The value should be within the criteria specified for the project. If measurement value drifted outside the criteria, the data will need to be qualified.

If the quality assurance project plan or the sampling and analysis plan do not list the drift criteria or the post-calibration criteria, use the criteria below.

Measurement	Post Calibration Criteria
Dissolved Oxygen	± 0.5 mg/L of sat. value* < 0.5 mg/L for the 0 mg/L solution, but not a negative value
Specific Conductance	$\pm 5\%$ of standard or ± 10 $\mu\text{S}/\text{cm}$ (whichever is greater)
pH	± 0.3 pH unit with pH 7 buffer*
Turbidity	$\pm 5\%$ of standard
ORP	± 10 mv*

Note: * Table 8.1, USEPA Region 1 YSI 6-Series Sondes and Data Logger SOP, January 30, 2007, revision 9.

7.0 DATA MANAGEMENT AND RECORDS MANAGEMENT

All calibration records must be documented in the project's log book or on a calibration log sheet. At a minimum, include the instrument manufacturer, model number, instrument identification number (when more than one instrument of the same model is used), the standards used to calibrate the instruments (including source), the calibration date, the instrument readings, the post calibration check, and the name of the person(s) who performed the calibration. An example of a calibration log sheet is shown in Table 1.

8.0 References

Standard Methods for the Examination of Water and Wastewater, 20th edition, 1998.

Methods for Chemical Analysis of Water and Wastes, EPA-600/4-79-020, Revised March 1983.

Turbidity - Methods for the Determination of Inorganic Substances in Environmental Samples, EPA/600/R-93/100, August 1993.

USEPA Region 1 YSI 6-Series Sondes and Data Logger SOP, January 30, 2007, revision 9.

USGS Guidelines and Standard Procedures for Continuous Water-Quality Monitors: Station Operation, Record Computation, and Data Reporting, Techniques and Methods 1-D3.

TABLE 1
INSTRUMENT CALIBRATION LOG

Project Name_____ Date_____

Weather_____

Calibrated by_____ Instrument_____

Serial Number_____

Parameters	Morning Calibration	Morning Temperature	End of Day Calibration Check*	End of Day Temperature
Specific Conductance Standard #1				
Specific Conductance Standard #2				
pH (7)				
pH (4)				
pH (10)				
ORP Zobel solution				
Dissolved Oxygen 100% water saturated air mg/L				
Dissolved Oxygen Zero Dissolved Oxygen Solution mg/L				
Barometric Pressure mm Hg		NA		NA
Turbidity Standard #1				
Turbidity Standard #2				
Turbidity Standard #3				

* For each Parameter, chose one standard as your check standard.
If possible, choose the one that is closest to the ambient measurement value.

Oxygen Solubility at Indicated Pressure

Temp. °C	Pressure (Hg)							mm in
	760	755	750	745	740	735	730	
0	29.92	29.72	29.53	29.33	29.13	28.94	28.74	mg/l
1	14.57	14.47	14.38	14.28	14.18	14.09	13.99	
2	14.17	14.08	13.98	13.89	13.79	13.70	13.61	
3	13.79	13.70	13.61	13.52	13.42	13.33	13.24	
4	13.43	13.34	13.25	13.16	13.07	12.98	12.90	
5	13.08	12.99	12.91	12.82	12.73	12.65	12.56	
6	12.74	12.66	12.57	12.49	12.40	12.32	12.23	
7	12.42	12.34	12.26	12.17	12.09	12.01	11.93	
8	12.11	12.03	11.95	11.87	11.79	11.71	11.63	
9	11.81	11.73	11.65	11.57	11.50	11.42	11.34	
10	11.53	11.45	11.38	11.30	11.22	11.15	11.07	
11	11.28	11.19	11.11	11.04	10.96	10.89	10.81	
12	10.99	10.92	10.84	10.77	10.70	10.62	10.55	
13	10.74	10.67	10.60	10.53	10.45	10.38	10.31	
14	10.50	10.43	10.36	10.29	10.22	10.15	10.08	
15	10.27	10.20	10.13	10.06	10.00	9.93	9.86	
16	10.05	9.98	9.92	9.85	9.78	9.71	9.65	
17	9.83	9.76	9.70	9.63	9.57	9.50	9.43	
18	9.63	9.57	9.50	9.44	9.37	9.31	9.24	
19	9.43	9.37	9.30	9.24	9.18	9.11	9.05	
20	9.24	9.18	9.12	9.05	8.99	8.93	8.87	
21	9.06	9.00	8.94	8.88	8.82	8.75	8.69	
22	8.88	8.82	8.76	8.70	8.64	8.58	8.52	
23	8.71	8.65	8.59	8.53	8.47	8.42	8.36	
24	8.55	8.49	8.43	8.38	8.32	8.26	8.20	
25	8.39	8.33	8.28	8.22	8.16	8.11	8.05	
26	8.24	8.18	8.13	8.07	8.02	7.96	7.90	
27	8.09	8.03	7.98	7.92	7.87	7.81	7.76	
28	7.95	7.90	7.84	7.79	7.73	7.68	7.62	
29	7.81	7.76	7.70	7.65	7.60	7.54	7.49	
30	7.68	7.63	7.57	7.52	7.47	7.42	7.36	
31	7.55	7.50	7.45	7.39	7.34	7.29	7.24	
32	7.42	7.37	7.32	7.27	7.22	7.16	7.11	
33	7.30	7.25	7.20	7.15	7.10	7.05	7.00	
34	7.08	7.13	7.08	7.03	6.98	6.93	6.88	
35	7.07	7.02	6.97	6.92	6.87	6.82	6.78	
36	6.95	6.90	6.85	6.80	6.76	6.71	6.66	
37	6.84	6.79	6.76	6.70	6.65	6.60	6.55	
38	6.73	6.68	6.64	6.59	6.54	6.49	6.45	
39	6.63	6.58	6.54	6.49	6.44	6.40	6.35	
40	6.52	6.47	6.43	6.38	6.35	6.29	6.24	
41	6.42	6.37	6.33	6.28	6.24	6.19	6.15	
42	6.32	6.27	6.23	6.18	6.14	6.09	6.05	
43	6.22	6.18	6.13	6.09	6.04	6.00	5.95	
44	6.13	6.09	6.04	6.00	5.95	5.91	5.87	
45	6.03	5.99	5.94	5.90	5.86	5.81	5.77	
	5.94	5.90	5.85	5.81	5.77	5.72	5.68	

(Continued)

Source: Draft EPA Handbook of Methods for Acid Deposition Studies, Field Operations for Surface Water Chemistry, EPA/600/4-89/020, August 1989.

Oxygen Solubility at Indicated Pressure (continued)

Temp. °C	Pressure (Hg)								mm in
	725	720	715	710	705	700	695	690	
0	28.54	28.35	28.15	27.95	27.76	27.56	27.36	27.17	mg/l
1	13.89	13.80	13.70	13.61	13.51	13.41	13.32	13.22	
2	13.51	13.42	13.33	13.23	13.14	13.04	12.95	12.86	
3	13.15	13.06	12.97	12.88	12.79	12.69	12.60	12.51	
4	12.81	12.72	12.63	12.54	12.45	12.36	12.27	12.18	
5	12.47	12.39	12.30	12.21	12.13	12.04	11.95	11.87	
6	12.15	12.06	11.98	11.89	11.81	11.73	11.64	11.56	
7	11.84	11.73	11.68	11.60	11.51	11.43	11.35	11.27	
8	11.55	11.47	11.39	11.31	11.22	11.14	11.06	10.98	
9	11.26	11.18	11.10	11.02	10.95	10.87	10.79	10.71	
10	10.99	10.92	10.84	10.76	10.69	10.61	10.53	10.46	
11	10.74	10.66	10.59	10.51	10.44	10.36	10.29	10.21	
12	10.48	10.40	10.33	10.28	10.18	10.11	10.04	9.96	
13	10.24	10.17	10.10	10.02	9.95	9.88	9.81	9.74	
14	10.01	9.94	9.87	9.80	9.73	9.66	9.59	9.52	
15	9.79	9.72	9.65	9.58	9.51	9.45	9.38	9.31	
16	9.58	9.51	9.44	9.38	9.31	9.24	9.18	9.11	
17	9.37	9.30	9.24	9.17	9.11	9.04	8.97	8.91	
18	9.18	9.11	9.05	8.98	8.92	8.85	8.79	8.73	
19	8.99	8.92	8.86	8.80	8.73	8.67	8.61	8.54	
20	8.81	8.74	8.68	8.62	8.56	8.49	8.43	8.37	
21	8.63	8.57	8.51	8.45	8.39	8.33	8.27	8.21	
22	8.46	8.40	8.34	8.28	8.22	8.16	8.10	8.04	
23	8.30	8.24	8.18	8.12	8.06	8.00	7.95	7.89	
24	8.15	8.09	8.03	7.97	7.91	7.86	7.80	7.74	
25	7.99	7.94	7.88	7.82	7.76	7.71	7.65	7.59	
26	7.85	7.79	7.74	7.68	7.60	7.57	7.51	7.46	
27	7.70	7.65	7.59	7.54	7.48	7.43	7.37	7.32	
28	7.57	7.52	7.46	7.41	7.35	7.30	7.25	7.19	
29	7.44	7.38	7.33	7.28	7.22	7.17	7.12	7.06	
30	7.31	7.26	7.21	7.15	7.10	7.05	7.00	6.94	
31	7.19	7.14	7.08	7.03	6.98	6.93	6.88	6.82	
32	7.06	7.01	6.96	6.91	6.86	6.81	6.76	6.70	
33	6.95	6.90	6.85	6.80	6.70	6.70	6.64	6.59	
34	6.83	6.78	6.73	6.68	6.63	6.58	6.53	6.48	
35	6.73	6.68	6.63	6.58	6.53	6.48	6.43	6.38	
36	6.61	6.56	6.51	6.47	6.42	6.37	6.36	6.27	
37	6.51	6.46	6.41	6.36	6.31	6.27	6.22	6.17	
38	6.40	6.35	6.31	6.26	6.21	6.16	6.12	6.07	
39	6.30	6.26	6.21	6.16	6.12	6.07	6.02	5.98	
40	6.26	6.15	6.11	6.06	6.01	5.97	5.92	5.87	
41	6.10	6.06	6.01	5.96	5.92	5.86	5.83	5.78	
42	6.00	5.96	5.91	5.87	5.82	5.78	5.73	5.69	
43	5.91	5.86	5.82	5.77	5.73	5.69	5.64	5.60	
44	5.82	5.78	5.73	5.69	5.65	5.60	5.56	5.51	
45	5.72	5.68	5.64	5.59	5.55	5.51	5.46	5.42	
	5.64	5.59	5.55	5.51	5.47	5.42	5.38	5.34	

Source: Draft EPA Handbook of Methods for Acid Deposition Studies, Field Operations for Surface Water Chemistry, EPA/600/4-89/020, August 1989.